

Design file:

```
module password_lock(  
    input clk,  
    input reset,  
    input [3:0] entered_password,  
    output reg access_granted,  
    output reg locked  
);  
parameter PASSWORD = 4'b1011;  
reg [1:0] wrong_attempts;  
always @(posedge clk or posedge reset)  
begin  
    if(reset)  
    begin  
        access_granted <= 0;  
        locked <= 0;  
        wrong_attempts <= 0;  
    end  
    else  
    begin  
        if(locked)  
        begin  
            access_granted <= 0;  
        end  
        else  
        begin  
            if(entered_password == PASSWORD)  
            begin  
                access_granted <= 1;  
                wrong_attempts <= 0;  
            end  
            else  
            begin  
                access_granted <= 0;  
                if(wrong_attempts == 2)  
                begin  
                    locked <= 1;  
                end  
                else  
                begin  
                    wrong_attempts <= wrong_attempts + 1;  
                end  
            end  
        end  
    end  
end  
end endmodule
```